# Color Theory

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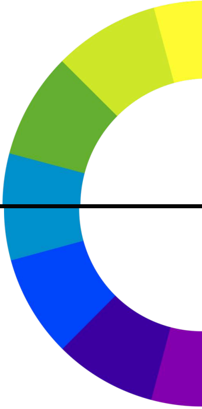

## What we will learn today

1. Why Color Matters in Design?
2. RGB and HEX
3. Warm and Cold Colors
4. Tints and Shades
5. HSL and HSB
6. Color Wheel
7. UI Color Palettes
8. UI Color Palettes Figma Activity : Generate color styles, Create a Simple UI with a Color Scheme

# Why Color Matters in Design?

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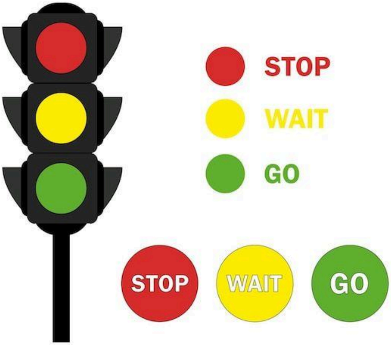
STOP
WAIT
GO
STOP
WAIT
GO

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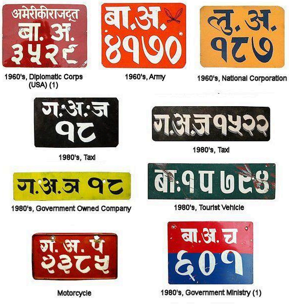
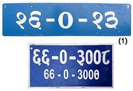
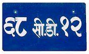

अमेरीकीराजदूत
बा. अ.
३५२९
1960's, Diplomatic Corps (USA) (1)
बा.अ.
४१७०
1960's, Army
लु. अ.
१८७
1960's, National Corporation
ग.अ.ज
१८
1980's, Taxi
ग.अ.ज१५२२
1980's, Taxi
ग.अ.ञ १८
1980's, Government Owned Company
बा:१प ७९४
1980's, Tourist Vehicle
ग. अ. पं
२३८५
Motorcycle
बा.अ.च
६०१
1980's, Government Ministry (1)


२६-०-२३
६६-०-३००८
66 - 0 - 3008

(1)

1980's, Diplomatic Corps/Aid Personnel



1980's, Diplomatic Corps

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4/2019

## Why Color Matters in Design?

Color is more than decoration – it's a powerful tool to guide users, set mood, and influence perception.

Key Points:

- Color shapes first impressions
- Drives emotions and brand recognition
- Guides users through interfaces
- Example: Red = Stop, Blue = Trust, Green = Go

# RGB and HEX

## RGB (Red, Green, Blue)

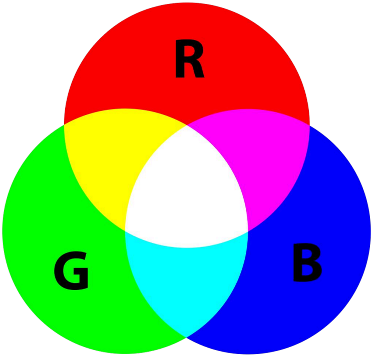
RGB is a color model used for digital screens where colors are created by combining different intensities of Red, Green, and Blue.

Key Points:

- rgb(255, 0, 0) → Pure Red
- rgb(0, 255, 0) → Pure Green
- rgb(0, 0, 255) → Pure Blue

Range:

- Each color has a value between 0 (none) and 255 (full intensity)
- rgb(255, 255, 255) = White
- rgb(0, 0, 0) = Black



## HEX (Hexadecimal Color Code)

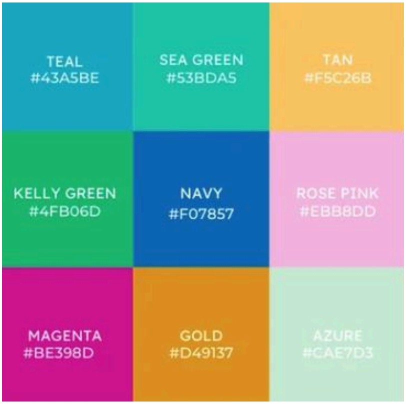
HEX is a shorthand for RGB values written in hexadecimal (base-16) format, commonly used in web design and CSS.

Examples:

- #FF0000 → Red
- #00FF00 → Green
- #0000FF → Blue
- #FFFFFF → White
- #000000 → Black

Structure:

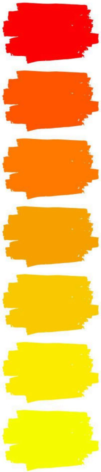
- 2 digits for Red, 2 for Green, 2 for Blue
- #RRGGBB



| | | |
|---|---|---|
| TEAL<br>#43A5BE | SEA GREEN<br>#53BDA5 | TAN<br>#F5C26B |
| KELLY GREEN<br>#4FB06D | NAVY<br>#F07857 | ROSE PINK<br>#EBB8DD |
| MAGENTA<br>#BE398D | GOLD<br>#D49137 | AZURE<br>#CAE7D3 |

# Warm and Cold Colors

## Warm Colors



Warm colors create feelings of energy, excitement, and comfort. They are often used to grab attention or create emotional warmth.

Examples:

- 🔴 Red – Passion, urgency, excitement
- 🟠 Orange – Creativity, enthusiasm
- 🟡 Yellow – Happiness, optimism, caution

Usage in UI:

- Call-to-action buttons
- Sale or urgency indicators
- Nºtifications and alerts

## Cool (Cold) Colors

Cool colors create a sense of calm, trust, and professionalism. They are often used for relaxing and stable designs.

Examples:

- 🔵 Blue – Trust, calmness, professionalism
- 🟢 Green – Growth, health, safety
- 🟣 Purple – Luxury, creativity, mystery

Usage in UI:

- Backgrounds for long reading
- Apps focused on productivity, finance, healthcare
- Success indicators (green checkmarks, confirmations)

# Tint and Shades

## Tints

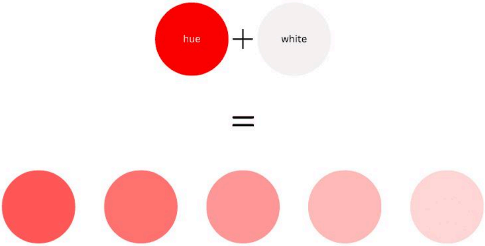
A Tint is created by adding white to a color, making it lighter and softer.

Examples:

- Red + White = Pink
- Blue + White = Light Blue

Usage in UI:

- Backgrounds
- Hover states



- Soft, minimal designs

## Shades

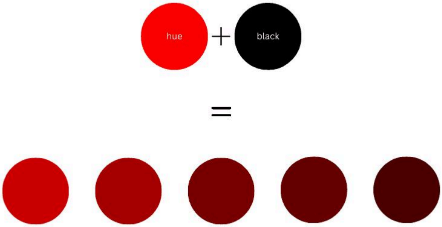
A Shade is created by adding black to a color, making it darker.

Examples:

- Red + Black = Dark Red (Maroon)
- Blue + Black = Navy Blue

Usage in UI:

- Buttons
- Text
- Dark mode backgrounds



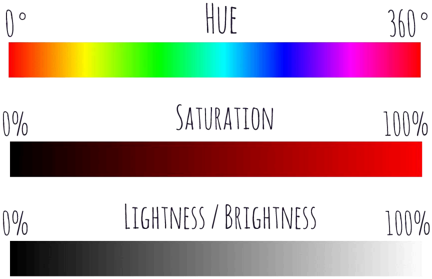
# HSL and HSB

## HSL (Hue, Saturation, Lightness)

HSL describes colors based on their tone, intensity, and brightness – it’s an intuitive way to adjust color appearance.

Breakdown:

- Hue (H): The type of color (0° to 360° on color wheel)
    - 0° = Red, 120° = Green, 240° = Blue
- Saturation (S): Intensity or purity of the color (0% to 100%)
    - 0% = Grey, 100% = Full color
- Lightness (L): Brightness of the color (0% = Black, 100% = White)


0°
HUE
360°
0%
SATURATION
100%
0%
LIGHTNESS / BRIGHTNESS
100%

## HSB (Hue, Saturation, Brightness)

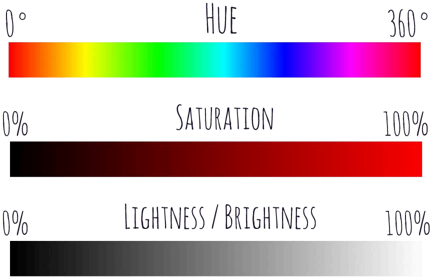
HSB is similar to HSL but uses Brightness instead of Lightness – often used in design tools for more visual control.

Breakdown:

- Hue (H): Same as HSL (color tone)
- Saturation (S): Color intensity
- Brightness (B): How light or dark the color appears

Usage in UI:

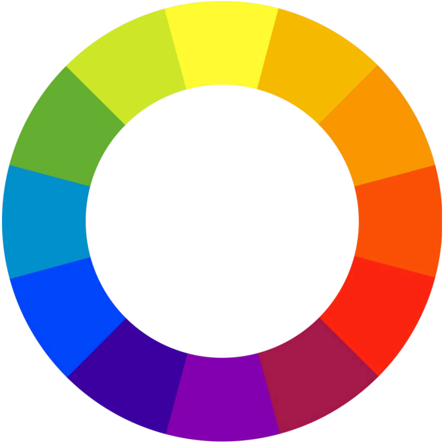
- Buttons
- Text
- Dark mode backgrounds

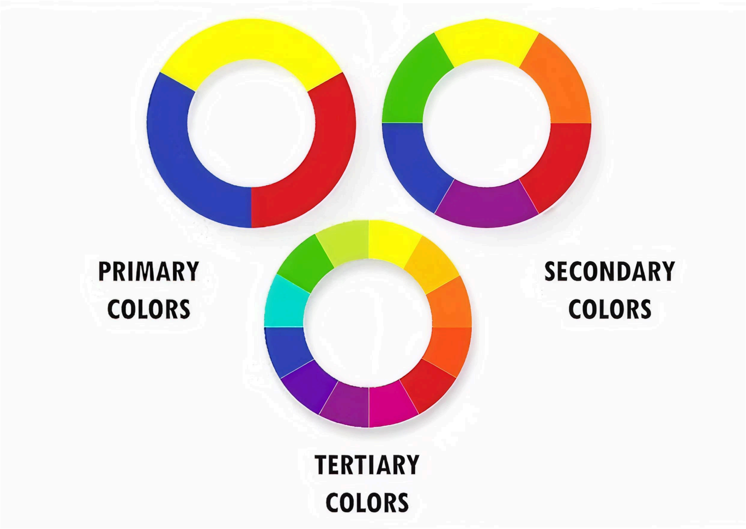

0°
HUE
360°
0%
SATURATION
100%
0%
LIGHTNESS / BRIGHTNESS
100%

# Color Wheel

## What is the Color Wheel?

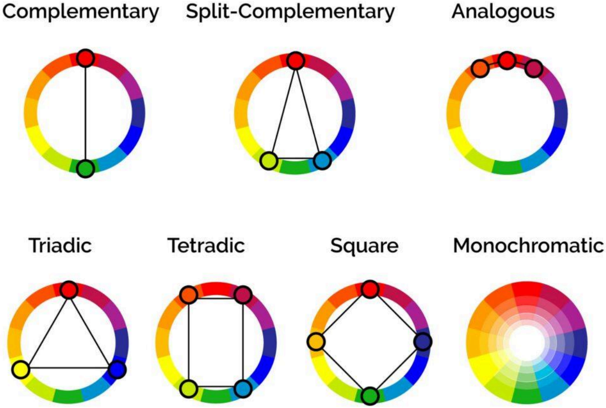
- The Color Wheel is a circular diagram that shows how colors relate to each other.
- Consists of 12 Colors
- Help to make decisions about color and acheive specific design goals




PRIMARY COLORS
SECONDARY COLORS
TERTIARY COLORS

## Color Schemes

- [https://www.canva.com/colors/color-wheel/](https://www.canva.com/colors/color-wheel/)
- [https://color.adobe.com/create/color-wheel](https://color.adobe.com/create/color-wheel)


Complementary
Split-Complementary
Analogous
Triadic
Tetradic
Square
Monochromatic

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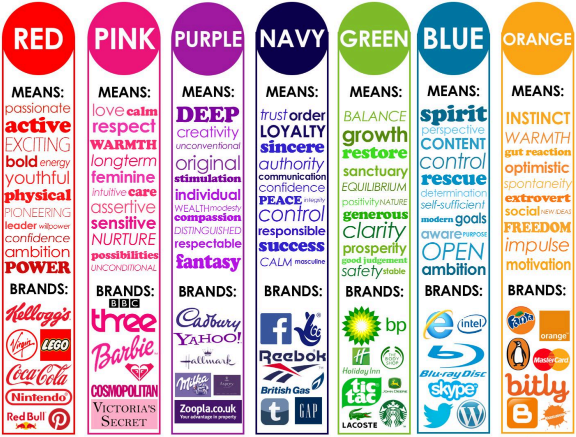
# Color Psychology

Color psychology is the phenomenon where color influences human behavior, emotions, and perceptions.



## RED

MEANS:

passionate
active
EXCITING
bold energy
youthful
physical
PIONEERING
leader willpower
confidence
ambition
POWER

BRANDS:

Kellogg's
Virgin
LEGO
Coca-Cola
Nintendo
Red Bull

## PINK

MEANS:

love calm
respect
WARMTH
longterm
feminine
intuitive care
assertive
sensitive
NURTURE
possibilities
UNCONDITIONAL

BRANDS:

BBC
three
Barbie
COSMOPOLITAN
VICTORIA'S SECRET

## PURPLE

MEANS:

DEEP
creativity
unconventional
original
stimulation
individual
WEALTH modesty
compassion
DISTINGUISHED
respectable
fantasy

BRANDS:

Cadbury
YAHOO!
Hallmark
Milka
Asprey
Zoopla.co.uk
Your advantage in property

## NAVY

MEANS:

trust order
LOYALTY
sincere
authority
communication
confidence
PEACE integrity
control
responsible
success
CALM masculine

BRANDS:

Reebok
British Gas
GAP

## GREEN

MEANS:

BALANCE
growth
restore
sanctuary
EQUILIBRIUM
positivity NATURE
generous
clarity
prosperity
good judgement
safety stable

BRANDS:

bp
Holiday Inn
THE BODY SHOP
tic tac
JOHN DEERE
LACOSTE

## BLUE

MEANS:

spirit
perspective
CONTENT
control
rescue
determination
self-sufficient
modern goals
aware PURPOSE
OPEN
ambition

BRANDS:

intel
Blu-ray Disc
skype

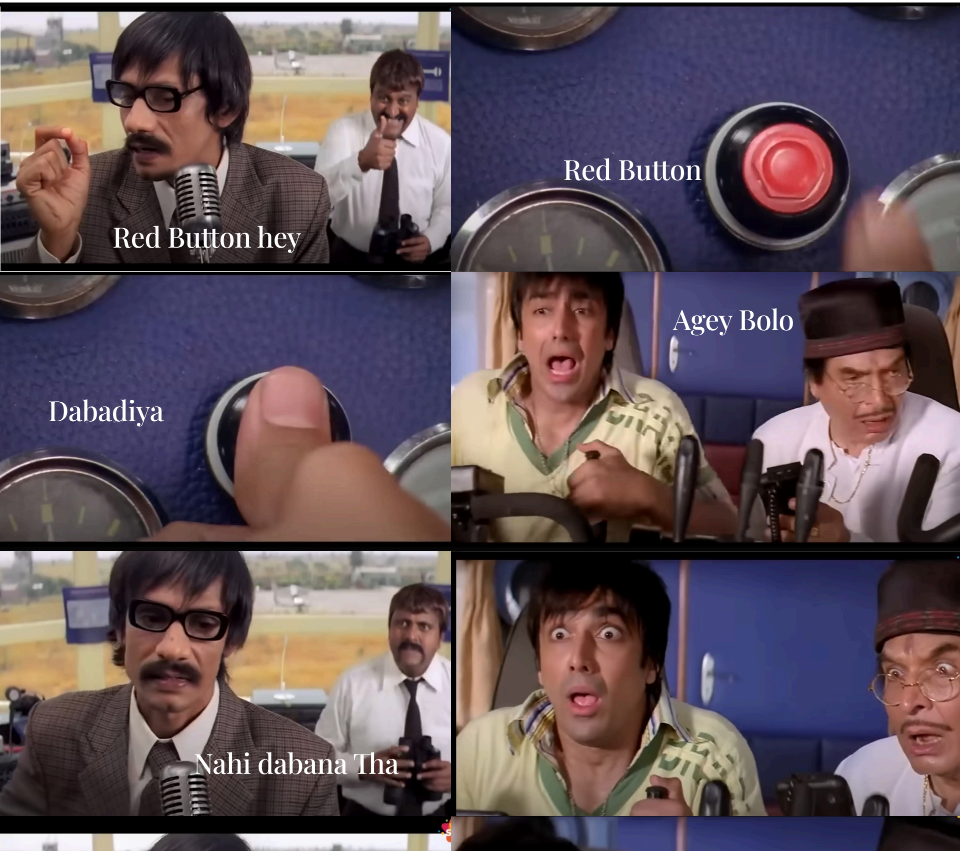
## ORANGE

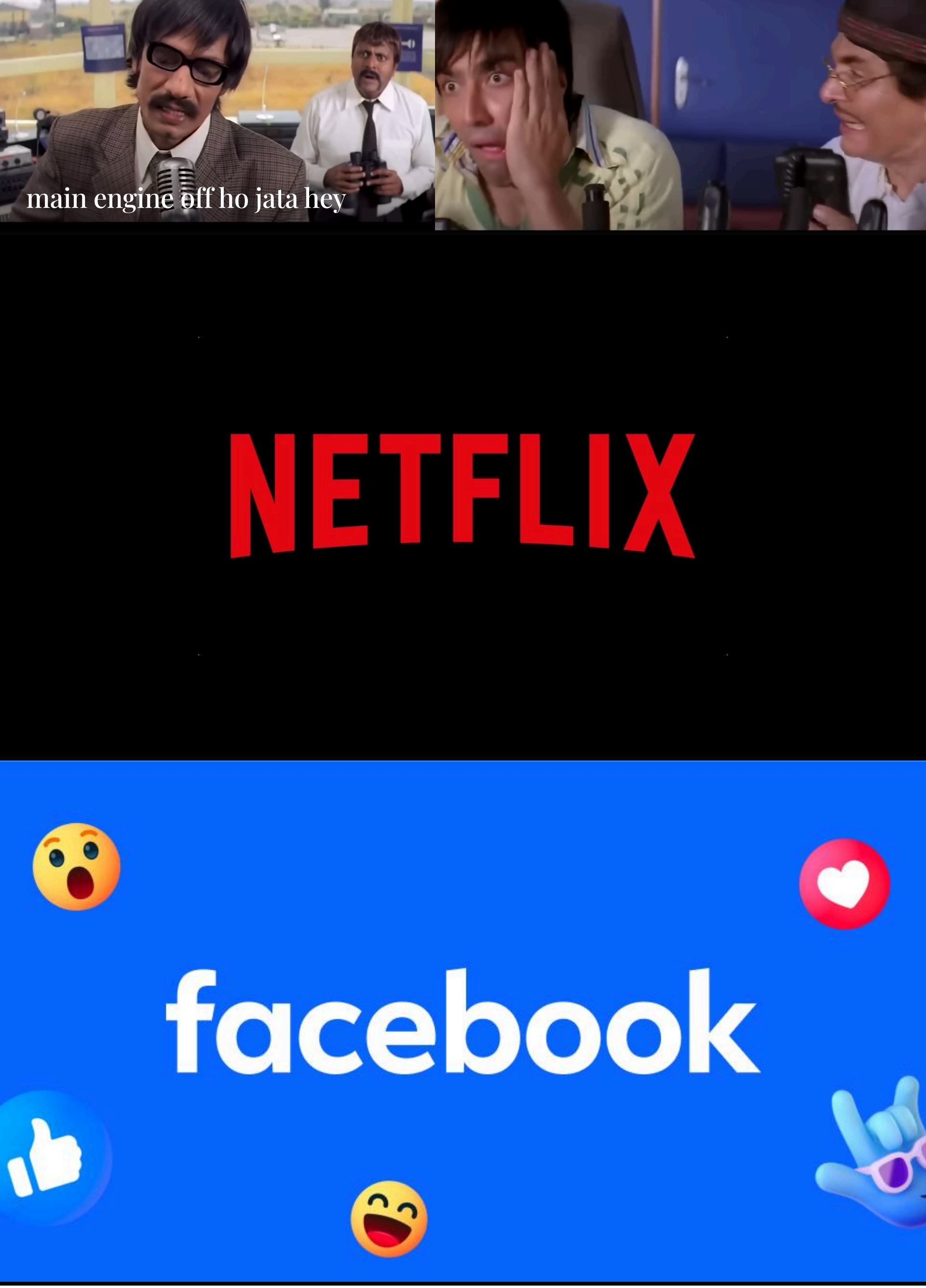
MEANS:

INSTINCT
WARMTH
gut reaction
optimistic
spontaneity
extrovert
social NEW IDEAS
FREEDOM
impulse
motivation

BRANDS:

Fanta
orange
MasterCard
bitly
nickelodeon


Red Button hey
Red Button
Dabadiya
Agey Bolo
Nahi dabana Tha


main engine off ho jata hey
NETFLIX
facebook

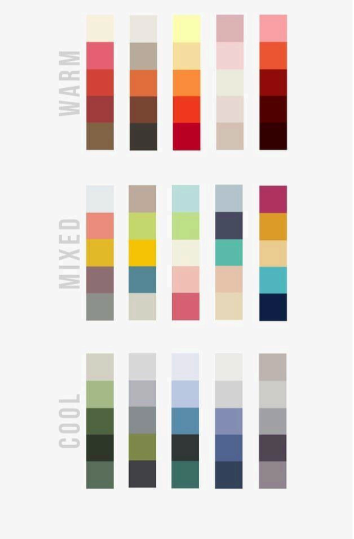
## Different Interpretations in different culture

- Red: Represents happiness in China and danger in the United States!
- Blue: Perceived as peaceful or calming across cultures, while it has authoritative and conservative meanings in some parts of the West!
- White: Symbolizes purity and innocence in some parts of the West and death or the supernatural in some parts of the East.
- Green: Represents growth, new life, and nature in numerous cultures, while it's sometimes connected to sickness or greed in others.

# UI Color Palettes

## UI Color Palettes

- A UI color palette is a collection of colors used in designing a user interface.
- It is like a designer paint box.
- It maintains visual harmony, defines brand identity, and enhances the user experience as it guides users seamlessly through the interface.


WARM
MIXED
COOL

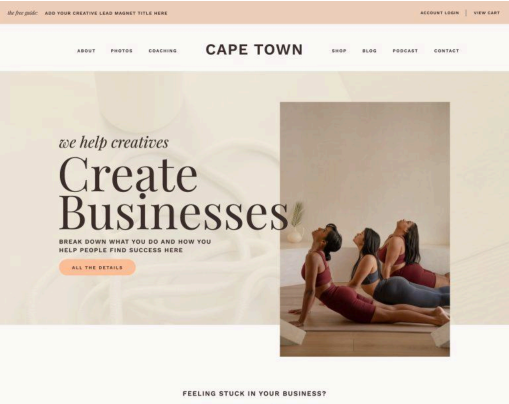
---

## Some tools for Building UI Color Palettes

- Adobe Color
- Coolors
- Color Hunt
- Colormind
- Paletton

# Examples

Color of
the Year
2024
PANTONE®
Peach Fuzz
3-1023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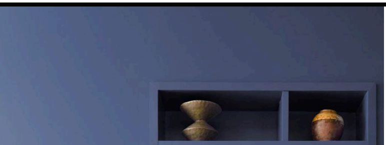

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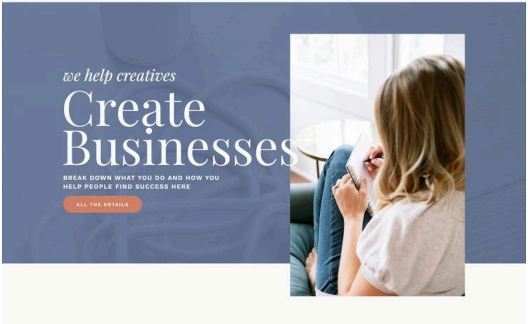

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ALL THE DETAILS

## Tips for choosing right colors

- Choose your primary color associated to your brand
- Avoid absolute white or black
- Maintain consistency
- Research your audience
- Consider color psychology
- Research and look out for inspirations
- Follow the 60-30-10 Rule
- Use Contrast Effectively and check the accessibility